

Omkar Kottawar

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SUMMARY

Computer Science undergraduate with experience building backend systems, Java applications, and AI-powered software through open-source contributions and end-to-end projects. Strong foundation in data structures, algorithms, object-oriented programming, and software engineering, with a focus on designing clean, modular, and maintainable systems. Passionate about solving challenging engineering problems and building reliable software at scale.

EDUCATION

VIT Bhopal University <i>B.Tech. in Computer Science and Engineering</i> - CGPA: 9.01/10	Sehore, MP Sep 2023 – May 2027
Datta Meghe World Academy <i>Higher Secondary Certificate</i> - Percentage: 81.4%	Airoli, MH Aug 2023
Datta Meghe World Academy <i>Senior Secondary Certificate</i> - Percentage: 95%	Airoli, MH Aug 2021

EXPERIENCE

Software Engineering Intern <i>FOSSEE, IIT Bombay</i>	Feb 2026 – May 2026 Mumbai, MH
- Refactored CAD generation modules within the OSDAG codebase, improving modularity, readability, and long-term maintainability. - Debugged complex software workflows, resolved generation inconsistencies, and improved reliability across multiple structural design configurations. - Collaborated using Git-based development workflows while enhancing documentation and reducing onboarding effort for future contributors.	

PROJECTS

MiniDB — Relational Database Engine <i>Java, Swing, OOP</i>	2026
- Designed and implemented an in-memory relational database supporting schema creation, CRUD operations, and constraint enforcement using object-oriented design principles. - Implemented primary-key uniqueness, NOT NULL validation, and type checking while maintaining schema integrity through a modular architecture. - Built an interactive Java Swing desktop application for database creation, table management, record insertion, updates, deletion, and real-time data visualization. - Structured the system using reusable components and separation of concerns, enabling future extensibility for SQL parsing and persistent storage.	
Context Handoff Protocol — AI-Powered Team Handoff Briefs <i>Python, FastAPI, HDBSCAN, LLMs</i>	2026
- Built a full-stack pipeline that ingests cross-platform team signals (Slack, GitHub, Notion, email) and synthesizes structured, AI-generated handoff briefs for engineering and product teams. - Engineered a modular four-stage pipeline (ingestion, semantic clustering via HDBSCAN, staleness scoring, LLM synthesis) with provider-agnostic embedding and LLM interfaces swappable via configuration alone. - Designed a FastAPI backend with async SQLAlchemy/SQLite persistence and a responsive single-page frontend, deployed on HuggingFace Spaces with pluggable connector architecture for real and synthetic data sources.	
ReviewSense — Customer Review Analytics Platform <i>Python, Pandas</i>	2025
- Built a review analytics application that processes customer reviews and generates interpretable satisfaction scores through a modular processing pipeline. - Implemented review ingestion, feature extraction, scoring, and CSV export with emphasis on reusable software components and maintainable code. - Designed the application for extensibility, enabling future integration with dashboards and large-scale datasets.	

TECHNICAL SKILLS

Programming: Java, Python, SQL
Backend: FastAPI, REST APIs, SQLAlchemy, Async Programming
Databases: SQLite, Relational Database Design
AI/ML: PyTorch, NumPy, Pandas, HDBSCAN, LLM Integration
Tools & Platforms: Git, Java Swing, Streamlit, Hugging Face Spaces
Cloud: Oracle Cloud Infrastructure

CERTIFICATIONS & TRAINING

Oracle Cloud Infrastructure AI Foundations Associate , Oracle University (2025–2027)
The Bits and Bytes of Computer Networking , Google (2025)
Machine Learning Specialization , DeepLearning.AI & Stanford Online (2026)
Retrieval-Augmented Generation (RAG) , DeepLearning.AI (2026)